ASSIGNMENT – 5

Introduction to Asymmetric Cryptography | BCA309T

Q1. Explain asymmetric cryptography, key roles, and security services.

Asymmetric cryptography uses a public key that can be shared and a private key kept secret. For confidentiality, Alice encrypts with Bob's public key and Bob decrypts with his private key. For authentication and non-repudiation, Alice signs with her private key and others verify using her public key.

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Alice → Encrypt with Bob's public key → Ciphertext → Decrypt with Bob's private key → Bob
Signature: Alice signs with her private key → Others verify with Alice's public key
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Q2. Compare symmetric and asymmetric cryptography.

Comparison table:

Heading	Symmetric	Asymmetric
Keys	One shared secret key	Public and private key pair
Distribution	Must share secret securely	Public key can be shared openly
Speed	Fast	Slower
Key length	Usually shorter	Usually longer for comparable security
Scalability	Many users need many shared keys	Easier to scale with public keys
Typical uses	Bulk data encryption	Key exchange, signatures, authentication

Q3. Explain trapdoor functions, Diffie-Hellman, digital signatures, and PKI.

A trapdoor one-way function is easy to calculate but difficult to reverse without secret information. Diffie-Hellman lets two parties create a shared secret over a public channel. Digital signatures use a private key to sign and a public key to verify a message. PKI uses trusted certificate authorities and digital certificates to link identities with public keys.

Q4. Trace the history of public-key cryptography.

Secret-key systems created a key-distribution problem because users had to share a secret securely. Diffie and Hellman introduced public-key ideas in 1976. Rivest, Shamir, and Adleman developed RSA in 1977. Earlier classified work at GCHQ by Ellis, Cocks, and Williamson explored related ideas. Later methods included ElGamal and Elliptic Curve Cryptography (ECC).

Q5. Give advantages and disadvantages. Why use hybrid cryptography?

Asymmetric cryptography helps with key distribution, authentication, and digital signatures, and scales well. It is slower than symmetric encryption. TLS therefore uses asymmetric techniques to authenticate or establish session keys, then fast symmetric encryption such as AES to protect the actual data.

Q6. Explain RSA key generation, encryption, and decryption.

1. Choose two large primes p and q.
2. Compute $n = p \times q$.
3. Compute $\varphi(n) = (p - 1)(q - 1)$.
4. Choose e such that $\gcd(e, \varphi(n)) = 1$.
5. Find d such that $d \times e \equiv 1 \pmod{\varphi(n)}$.

Public key: (e, n). Private key: (d, n).
Encryption: C = M^e mod n.
Decryption: M = C^d mod n.
Because d is the modular inverse of e, RSA's modular arithmetic recovers the original message. Real systems must use secure padding.

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Alice's message → RSA encryption with public key → Ciphertext → RSA decryption with private key → Original mess
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Q7. Explain RSA key size, factoring, ECC, and quantum risk.

RSA key size is the bit length of modulus n. Its security relies on the difficulty of factoring n into its prime factors. A 1024-bit key is too weak for new protection; 2048 bits is a common minimum, while 3072 or 4096 bits offer more security but require more computation. ECC offers comparable security with shorter keys. A sufficiently powerful quantum computer using Shor's algorithm could break RSA.

Q8. Discuss RSA attacks and good practices.

Risks include factoring attacks, small-exponent or small-message attacks, common-modulus misuse, timing/side-channel leakage, and chosen-ciphertext attacks. OAEP padding protects RSA encryption from unsafe textbook-RSA use. Digital signatures use a private key to sign and a public key to verify, with secure signature padding. Best practices: use at least 2048-bit keys, strong randomness, standard padding, constant-time implementations, trusted libraries, and secure private-key storage.